Component Specification

Hadamard Perturbation & Resistor Network for Analog Matrix 2T1C

This document summarizes the calculated and selected component values required to realize the digitally driven Hadamard perturbation (±15.6 mV) across the floating Gate node of the 2T1C cell. This is achieved via capacitive coupling using C_{PTU} and Return-to-Zero (RZ) Tri-state logic control.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Design Targets & Impedance

The total equivalent Thevenin impedance (R1 || R2 || Rseries) is **57.4 Ω**. This source resistance drives a total row capacitance of approximately **6 nF** (6 cells × 0.99 nF series equivalent). The resulting RC time constant (τ) is **0.344 μs**, ensuring a >99.6% settling accuracy within the 2 μs sampling window (satisfying the 1/255 LSB settling requirement).

3. Semiconductors & Control Logic (FPGA)

COMPONENT	VALUE / TYPE	DESCRIPTION / FUNCTION
Synaptic Transistor	DOX3134A	N-channel trench MOSFET, optimized for analog multiplication operating within the linear (triode) region.
Digital Driver	FPGA I/O	3.3V CMOS logic. Configured to **16 mA Drive Strength** to ensure sharp edges and minimal internal driver resistance. Must support Tri-state (High-Z) functionality.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.